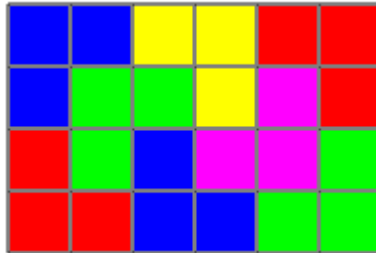


Filling Trominos

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to fill an $n \times m$ grid using L-trominos (three squares that have an L-shape). For example, here is one way to fill a 4×6 grid:



Input

The first input line has an integer t : the number of tests.

After that, there are t lines that describe the tests. Each line has two integers n and m .

Output

For each test, print YES if there is a solution, and NO otherwise.

If there is a solution, also print n lines that each contain m letters between A–Z. Adjacent squares must have the same letter exactly when they belong to the same tromino. You can print any valid solution.

Constraints

- $1 \leq t \leq 100$
- $1 \leq n, m \leq 100$

Example

Input	Output
2 4 6 4 7	YES AADDDBB ACCDEB BCAEEC BBAACC NO